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Distributed resilient adaptive consensus tracking control of nonlinear multi-agent systems dealing with deception attacks via K-filters approach☆

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ABSTRACT

It poses technical difficulty to achieve distributed consensus tracking control with input quantizations and deception attacks for nonlinear multi-agent systems subject to mismatched parametric uncertainties via backstepping design. The underlying problem becomes even more complicated because *i*) the system contains mismatched uncertainties; *ii*) the output becomes unavailable after attacks, making the conventional recursive control design strategy using traditional error surfaces unsuitable; *iii*) since only the compromised output signal is available, the impacts of nonlinear items related to unknown attack weights become difficult to be explicitly addressed in the existing available output feedback control results. In this paper, we present a solution to circumvent these obstacles by constructing a set of new K-filters using compromised output only, which, together with a coordinate transformation involving the attacked output and filter variables, allow a new form of backstepping based distributed resilient adaptive control protocol to be developed. Meanwhile, an extra compensation term is incorporated into the real controller to handle the effect of quantization on the system stability. It is shown that, with proper choices of Lyapunov functions, the global uniform boundedness of all the closed-loop signals and the asymptotically output consensus tracking can be achieved. A practice example is provided to verify the benefits of our scheme.

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1. Introduction

Consensus control of multi-agent systems(MASs) has become a rapidly emerging topic in various research communities over the past decades (Hu & Zheng, 2014; Li, Liu, Ren, & Xie, 2013; Wang, Wen, & Huang, 2017; Yiguang Hong, Jiangping Hu, & Linxin Gao, 2006) due to its increasing and wide application in areas ranging from natural networks to unmanned aerial vehicles. In the aforementioned literature, the network communication among agents is assumed to function normally without cyber-attacks. Recently, with the cross penetration of control, computing and communication technologies, cyber security problem has become so important that cannot be ignored in the field of control (An & Yang, 2019; Antsaklis, 2014; Li, Shi, Cheng, Chen, & Quevedo, 2015; Zhang, Deng, & Feng, 2022). In many practical situations, due to the transmission of data over open wireless networks, systems are easily vulnerable to malicious attacks from attackers. Typically, the types of cyberattacks can be broadly divided into three main groups: the denial-of-service (DOS) attacks (Deng, Zhang, & Feng, 2022) try to prevent the information transmission, the deception attacks (Yoo, 2019) try to inject the false information into the true data, and replay attacks (Li, Wang, Zou, Chen, & Yu, 2023) try to delay the information transmission. Among them, the deception attack is a malicious attack mode in the process of network transmission, which might influence steady-state performance of the systems, leading to instability. Therefore, it is a critical issue to develop resilient control methods to effectively deal with the impact of deception attacks. Until now, some meaningful resilient adaptive control results about the deception attacks have been devoted for linear systems (An & Yang, 2018; Jin, Haddad, & Yucelen, 2017). However, these adaptive resilient control approaches (An & Yang, 2018; Jin et al., 2017) cannot be applied to the more general case of high-order nonlinear MASs with mismatched parametric uncertainties.

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It is well known that the backstepping technique is an effective means to handle highly unknown and unmatched nonlinearities for nonlinear systems (Lin & Qian, 2002; Long, Wang, & Zhao, 2015). The important point is that for the security problems of nonlinear MASs suffering from malicious attacks, since the state signals are damaged by attacks, only the wrong states can be obtained and employed directly to design adaptive control schemes. Thus, the research on resilient adaptive backstepping control for the nonlinear MASs with deception attacks is a challenging task since the traditional error transformation technology mentioned in Lin and Qian (2002), Long et al. (2015) will result in the failure to complete the controller design objectives based on the backstepping and only the compromised states are available. More recently, some adaptive backstepping-based resilient control results against deception attacks were proposed for uncertain nonlinear systems (Lv, 2021; Ren & Yang, 2020; Yoo, 2019; Zhao & Yang, 2023). However, in the above works (Lv, 2021; Ren & Yang, 2020; Yoo, 2019; Zhao & Yang, 2023), the systems are all assumed that the full states are available, which restricts these schemes to be applied in practice. Focusing on the issue of output feedback control under deception attacks is undoubtedly meaningful although challenging. More recently, in Liu, Tang, Zhao, and Zhang (2021), Wang, Cao, Niu, and Song (2023), a novel state observer was constructed based on compromised output and a resilient adaptive controller was designed for nonlinear systems and MASs under deception attacks. However, the attack signal in Liu et al. (2021), Wang et al. (2023) was simply reduced to an unknown bounded constant by using inequality method, and the impact of attack term still exists. In other words, in the framework of output feedback control, results on networked nonlinear MASs with mismatched uncertainties under the sensor and actuator deception attacks are still limited and the related issues have not been well addressed. In addition, the control signals are usually quantized before transmission due to limited bandwidth. An important issue for networked MASs is to minimize the amount of information to be transmitted through communication channels while achieving satisfactory closed-loop performance. Many fundamental results on quantized control have been addressed in Corradini and Orlando (2008), Xing, Wen, Zhu, Su, and Liu (2016), Yu and Lin (2016). However, considering the cybersecurity problems encountered by the system in the process of signal transmission under deception attacks, the output measured by the sensor is corrupted, that is, the false output signal is applied to design controllers recursively, thus the quantized control input is based on false state variable, rendering those existing Lyapunov-stability-based resilient recursive design methods inapplicable.

Motivated by the above analysis and discussion, in this work, we aim to solve the problem of resilient adaptive quantized control for nonlinear MASs under the sensor and actuator deception attacks. The key contributions and features of our method can be summarized as follows:

(1) Different from An and Yang (2018), Gao, Huang, Su, and Zhao (2022), Jin et al. (2017), more general systems with mismatched parametric uncertainties are considered in the following terms. (i) The subsystems are nonlinear and allowed to have arbitrary relative degree and nonidentical dynamics; (ii) Intrinsic mismatched unknown parameters and input quantization are simultaneously involved. Unlike Lv (2021), Ren and Yang (2020), Yoo (2019), Zhao and Yang (2023), whose results can only be contingent upon the availability of complete state information, this paper employs compromised and inaccurate output, rendering the control design and stability analysis complicated;

(2) Although the authors in Liu et al. (2021), Wang et al. (2023) consider the observer based resilient control, the process of its observer design and stability analysis is rather complicated due to the impact of time-varying attack signal, particularly those related to the unknown attack terms are very challenging to handle. In contrast, with the aid of new K-filters, the malicious impact of the unknown deception attacks is compensated in this work, with which asymptotic stability is able to achieved;

(3) Due to the unknown deception attacks appearing in the output of each subsystem, the traditional error transformation of backstepping design procedure in Lin and Qian (2002), Long et al. (2015) are not available. By introducing error variables with the compromised output of the filters, a novel backstepping based distributed adaptive control protocol is proposed, and the additional information transmission of local control inputs or local neighborhood consensus errors among connected subsystems can be avoided. Furthermore, unlike Lv (2021), Ren and Yang (2020), Yoo (2019), Zhao and Yang (2023), this paper considers not only the sensor and actuator deception attacks but also input quantization. By employing an effective compensation term in the control law, the impacts of nonlinearities related to quantizer can be properly dealt with, making the concerned problem more practical yet technically more challenging.

2. Problem formulation and preliminaries

2.1. System model

In this part, only the output information after attacks can be used for controller design, and a group of N nonlinear subsystems are modeled with unknown deception attacks and input quantizations.

$$\begin{aligned}
&\dot{x}_i = A_i x_i + \phi_i(y_i)\theta_i + \psi_i(y_i) + \begin{bmatrix} 0_{(r_i-1)\times 1} \\ b_i \end{bmatrix} [q(u_i) + \rho_{i,a}(t, x_{i,1})] \\
&\check{x}_{i,1} = x_{i,1} + \omega_{i,s}(t, x_{i,1}) \\
&y_i = x_{i,1}, \quad \text{for } i = 1, \ldots, N
\end{aligned} \tag{1}$$

where $A_i = \begin{bmatrix} 0 & & \\ \vdots & I_{n_i-1} & \\ 0 & \ldots & 0 \end{bmatrix}$, $x_i = [x_{i,1}, \ldots, x_{i,n_i}]^T \in R^{n_i}$ stand for the system states, $y_i \in R$ and $u_i \in R$ are the system output and input, respectively. $\phi_i(y_i) \in R^{n_i \times p_i}$ and $\psi_i(y_i) \in R^n$ are the known smooth nonlinear functions. $\theta_i \in R^{p_i}$ are vectors of unknown parameters, $q(u_i)$ denotes the output of quantizer, $b_i = [b_{i,m_i}, \ldots, b_{i,0}]^T \in R^{m_i+1}$ are unknown constant vectors, and the sign of b_{i,m_i}, i.e., $\text{sign}(b_{i,m_i})$ is also unknown. $\rho_{i,a}$ is an actuator deception attack, $\omega_{i,s}(t, x_{i,1})$ is the sensor deception attack and $\check{x}_{i,1}$ is the available system state after attack.

Fig. 1 shows that the output and the quantizer of agent i, the information transmission of agent j ($j \in N_i$) is attacked by unknown deception attacks. In addition, although $\phi_i(\cdot) \in R^{n_i \times p_i}$ and $\psi_i(\cdot) \in R^{n_i}$ are known for the controller design, $\phi_i(y_i)$ and $\psi_i(y_i)$ become unknown since y_i is not available in the presence of cyberattacks, which brings additional difficulty in the process of observer and backstepping design, and further discussion will be held later.

2.2. Graph theory

An undirected graph $G \triangleq \{V, E\}$ is introduced to denote the interactions among the group of N subsystems, where $V =$

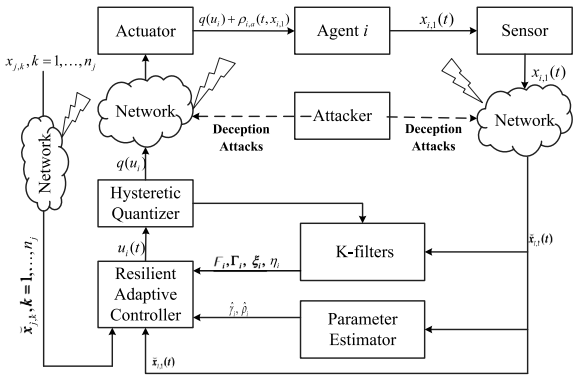


Fig. 1. The control block diagram for nonlinear MASs with deception attacks.

$\{1, \ldots, N\}$ represents a set of follower agents, N is the number of agents, $E \subseteq \{e_{ij} : i = 1, \ldots, N, j = 1, \ldots, N, i \neq j\}$ denotes a set of edges. Since G is undirected, if the edge $(i, j) \in E$, then $(j, i) \in E$. In this case, subsystem j is called a neighbor of subsystem i, and vice versa. $N_i = \{j \in V : (i, j) \in E\}$ represents all the neighbors of node i. The adjacency matrix is described as $A = [a_{ij}] \in R^{N\times N}$, if $(i, j) \in E$, $a_{ij} = a_{ji} = 1$, and if $(i, j) \notin E$, $a_{ij} = a_{ji} = 0$. The Laplacian matrix is defined as $L = D - A$, where $L \in R^{N\times N}$, $D = \text{diag}(d_i)$ is the in-degree matrix of ith follower, and $d_i = \sum_{j\in N_i} a_{ij}$.

The objective of this paper is to determine appropriate state estimator and distributed resilient adaptive controller u_i for each follower agent (1) such that all the signals in the closed-loop system are bounded and the outputs of N subsystems can asymptotically track a desired trajectory y_d under deception attacks.

To achieve the above-mentioned control purpose, the following assumptions are needed.

Assumption 1. The undirected graph G is fixed and connected. The desired trajectory y_d is available to at least one subsystem in G.

Assumption 2. The first order derivative of $y_d(t)$ is bounded, piecewise continuous, there exists an unknown positive constant Y_d such that $|\dot{y}_d| \leq Y_d$.

Assumption 3. $B_i(s)$ is a Hurwitz polynomial, i.e., all zeros of the polynomial $b_{i,m_i}s^{m_i} + \cdots + b_{i,1}s + b_{i,0}$ are stable, and the relative degree $r_i(= n_i - m_i)$ is known.

Assumption 4 (*Yoo, 2019*)**.** The terms $\omega_{i,s}(t, x_{i,1})$ and $\rho_{i,a}(t, x_{i,1})$ caused by deception attacks are defined as $\omega_{i,s}(t, x_{i,1}) = \zeta(t)x_{i,1}(t)$ and $\rho_{i,a}(t, x_{i,1}) = w_a(t)\Delta(x_{i,1})$ with unknown time-varying signals $\zeta(t)(\zeta(t) \neq -1)$ and $w_a(t)$. $\Delta(x_{i,1})$ is a known smooth nonlinear function, and there exist unknown constants $\bar{\zeta}$, $\bar{\zeta}_d$ and $\bar{w}_a$ such that $|\zeta(t)| \leq \bar{\zeta}$, $|\dot{\zeta}(t)| \leq \bar{\zeta}_d$, $|w_a(t)| \leq \bar{w}_a$ and $|\dot{w}_a(t)| \leq \bar{w}_{ad}$.

Define $\lambda = (1 + \zeta)^{-1}$, by recalling Assumption 4, we have $\lambda_m \leq |\lambda| \leq \lambda_M$ and $|\dot{\lambda}| \leq \bar{\dot{\lambda}}$, where $\lambda_m, \lambda_M, \bar{\dot{\lambda}}$ are unknown positive constants.

Remark 1. In comparison with the existing adaptive secure control results in Yoo (2019), where the sign $1 + \zeta(t) > 0$ for the attack weight is needed, here the constraint condition on $\zeta(t)$ is relaxed in Assumption 4. Besides, from the description of system (1) and Assumption 4, we can get that $\breve{x}_{i,1} = (1 + \zeta(t))x_{i,1}$. Thus, the condition $1 + \zeta(t) \neq 0$ is to avoid the singularity problem, causing instability of the system.

In addition, the hysteretic quantizer considered in this paper is given by Zhou, Wen, and Yang (2013):

$$q(u_i) = \begin{cases} u_{i,j}\text{sgn}(u_i), & \frac{u_{i,j}}{1+\delta} < |u_i| \leq u_{i,j}, \dot{u}_i < 0, \text{ or} \\ & u_{i,j} < |u_i| \leq \frac{u_{i,j}}{1-\delta}, \dot{u}_i > 0, \\ u_{i,j}(1+\delta)\text{sgn}(u_i), & u_{i,j} < |u_i| \leq \frac{u_{i,j}}{1-\delta}, \dot{u}_i < 0, \text{ or} \\ & \frac{u_{i,j}}{1-\delta} < |u_i| \leq \frac{u_{i,j}(1+\delta)}{1-\delta}, \dot{u}_i > 0, \\ 0, & 0 \leq |u_i| < \frac{u_{\min}}{1+\delta}, \dot{u}_i < 0, \text{ or} \\ & \frac{u_{\min}}{1+\delta} \leq |u_i| < u_{\min}, \dot{u}_i > 0, \\ q(u_i(t^-)) & \dot{u}_i = 0 \end{cases} \tag{2}$$

where $u_{i,j} = \mu^{1-j}u_{i,\min}$, $j = 1, 2, \ldots$, and $u_{i,\min} > 0$ denotes the size of the dead-zone for $q(u_i)$ and $0 < \mu < 1$, $\delta = (1-\mu)/(1+\mu)$, μ is a measure of quantization density.

The hysteretic quantizer $q(u_i)$ can be decomposed into a linear part and a nonlinear part as follows:

$$q(u_i) = u_i + d_i. \tag{3}$$

Lemma 1. *The nonlinearity d_i satisfies the following inequalities:*

$$d_i^2 \leq \delta^2 u_i^2, \quad \forall |u_i| \geq u_{i,min}, \tag{4}$$

$$d_i^2 \leq u_{i,min}^2, \quad \forall |u_i| \leq u_{min}. \tag{5}$$

Proof. The proof is given in Zhou et al. (2013).

Lemma 2 (*Zhang, Song, & Wen, 2021*)**.** *Let $V(t)$ be smooth and radially unbounded function defined on $[0, \infty)$, $\forall t \geq 0$. Suppose that the following inequality holds:*

$$V(t) \leq D + \sum_{i=1}^{N} \int_0^t \left(\beta_i(t)N_i(\kappa_i) + 1\right)\dot{\kappa}_i d\tau \tag{6}$$

where D represents a suitable positive constant, and $\beta_i(t)$ is a time-varying parameter, which takes values in the unknown closed intervals $I = [l^-, l^+]$ with $0 \notin I$, then $V(t)$, $\kappa_i(t)$ and $\sum_{i=1}^{N} \int_0^t \left(\beta_i(t)N_i(\kappa_i) + 1\right)\dot{\kappa}_i d\tau$ must be bounded over $[0, \infty)$.

Lemma 3 (*Lin & Qian, 2002*)**.** *For any real-valued continuous function $s(x, y)$, there exist smooth functions $s_1(x) \geq 1$ and $s_2(y) \geq 1$ so that the inequality $|s(x, y)| \leq s_1(x)s_2(y)$ holds.*

Lemma 4. *For any given $\varpi \in R^m$, there exists a $\sigma > 0$, such that*

$$0 \leq \|\varpi\| - \frac{\|\varpi\|^2}{\sqrt{\|\varpi\|^2 + \sigma^2}} \leq \sigma. \tag{7}$$

Proof. Since $\sigma > 0$, we can obtain

$$\|\varpi\| = \frac{\|\varpi\|\sqrt{\|\varpi\|^2 + \sigma^2}}{\sqrt{\|\varpi\|^2 + \sigma^2}} \leq \frac{\|\varpi\|^2}{\sqrt{\|\varpi\|^2 + \sigma^2}} + \sigma. \tag{8}$$

The proof is completed.

3. The design of the resilient adaptive controller based on K-filters

3.1. K-filters and observer design

In this paper, only the output $x_{i,1}$ can be measured in system (1), therefore, a state observer needs to be developed to realize the estimation of unknown state $x_{i,k}$, $k = 2, \ldots, n_i$. However, unlike the general output feedback control method (Zhou, Shi, Lu, & Xu, 2011), the output of the system is corrupted due to the existence of deception attacks. Thus, only the false state $\breve{x}_{i,1}$

can be used to design the observer, making the observer design challenging.

In order to avoid the affect of unknown attack signal $\omega_{i,s}(t, x_{i,1})$ under deception attack, a set of new K-filters for estimating n_i-1 unmeasurable states $x_{i,2}, \ldots, x_{i,n_i}$ using false output $\check{x}_{i,1}$ only are constructed as follows:

$$\dot{\digamma}_i = A_{i,0}\digamma_i + H_{i,0}\check{x}_{i,1}, \digamma_i \in R^{n_i} \tag{9}$$

$$\dot{\Gamma}_i = A_{i,0}\Gamma_i + \phi_i(\check{x}_{i,1}), \Gamma_i \in R^{n_i \times p_i} \tag{10}$$

$$\dot{\xi}_i = A_{i,0}\xi_i + \psi_i(\check{x}_{i,1}), \xi_i \in R^{n_i} \tag{11}$$

$$\dot{\chi}_i = A_{i,0}\chi_i + e_{n_i,n_i}\Delta(\check{x}_{i,1}), \chi_i \in R^{n_i} \tag{12}$$

$$\dot{\nu}_i = A_{i,0}\nu_i + e_{n_i,n_i}q(u_i), \nu_i \in R^{n_i} \tag{13}$$

where $A_{i,0} = \begin{bmatrix} -k_{i,1} & \\ \vdots & I_{n_i-1} \\ -k_{i,n_i} & 0_{1\times(n_i-1)} \end{bmatrix} \in R^{n_i \times n_i}$ with $k_i = [k_{i,1}, \ldots, k_{i,n_i}]^T \in R^{n_i}$ is chosen such that the matrix $A_{i,0}$ is a strict Hurwitz matrix. $H_{i,0} = [1, 1, \ldots, 1]^T \in R^{n_i}$, $\phi_i(\cdot) = [\phi_{i,1}(\cdot), \ldots, \phi_{i,n_i}(\cdot)]^T \in R^{n_i \times p_i}$ and $\psi_i(\cdot) = [\psi_{i,1}(\cdot), \ldots, \psi_{i,n_i}(\cdot)]^T \in R^{n_i}$ and $e_{n_i,n_i} = [0, \ldots, 1]^T \in R^{n_i}$.

Using the mean value theory (Hassan et al., 2002), it naturally holds that $\phi_i(\check{x}_{i,1}) = \phi_i(x_{i,1}) + o_{i,1}(1-\lambda)H_{i,0}\check{x}_{i,1}$, $\psi_i(\check{x}_{i,1}) = \psi_i(x_{i,1}) + o_{i,2}(1-\lambda)H_{i,0}\check{x}_{i,1}$, $\Delta_i(\check{x}_{i,1}) = \Delta_i(x_{i,1}) + o_{i,3}(1-\lambda)H_{i,0}\check{x}_{i,1}$, where $o_{i,1} = \text{diag}\{o_{i,(1,j)}\}$ with $o_{i,(1,j)} = [o_{i,(1,j1)}, \ldots, o_{i,(1,jp_i)}] \in R^{p_i}$, $o_{i,2} = \text{diag}\{o_{i,(2,j)}\}$ and $o_{i,3} = [o_{i,(3,1)}, 0, \ldots, 0] \in R^{n_i}$ are all the unknown derivative, $j = 1, \ldots, n_i$.

Define $h_{i,k} = A_{i,0}^k\chi_i$, $\eta_{i,k} = A_{i,0}^k\nu_i$, $k = 0, 1, \ldots, m_i$, by using (8.22), (8.23) in Krstic, Kokotovic, and Kanellakopoulos (1995), we have

$$\dot{h}_{i,k} = A_{i,0}h_{i,k} + e_{n_i,n_i-k}\Delta(\check{x}_{i,1}) \tag{14}$$

$$\dot{\eta}_{i,k} = A_{i,0}\eta_{i,k} + e_{n_i,n_i-k}q(u_i) \tag{15}$$

where $A_{i,0}^j e_{n_i,n_i} = e_{n_i,n_i-j}, j = 0, 1, \ldots, n_i$. Based on these designed filters, we introduce the auxiliary variable

$$\tilde{x}_i = \Xi_i\digamma_i + \Gamma_i\theta_i + \xi_i + \sum_{k=0}^{m_i} b_{i,k}\eta_{i,k} + w_a(t)\sum_{k=0}^{m_i} b_{i,k}h_{i,k} \tag{16}$$

where $\tilde{x}_i = [\tilde{x}_{i,1}, \ldots, \tilde{x}_{i,n_i}]^T$, $\Xi_i = k_i\lambda - o_{i,1}\theta_i(1-\lambda) - o_{i,2}(1-\lambda) - \sum_{k=0}^{m_i} e_{n_i,n_i-k}b_{i,k}w_a(t)o_{i,3}(1-\lambda)$, and $k_i = \text{diag}\{k_{i,1}, \ldots, k_{i,n_i}\}$.

From (9)–(16), we can design an observer as

$$\begin{aligned}\dot{\hat{x}}_i =& A_{i,0}\tilde{x}_i + \begin{bmatrix} 0_{(r_i-1)\times 1} \\ b_i \end{bmatrix}[q(u_i) + w_a(t)\Delta(x_{i,1})] + k_i\lambda H_{i,0}\check{x}_{i,1} \\ &+ \dot{\Xi}_i\digamma_i + \dot{w}_a(t)\sum_{k=0}^{m_i} b_{i,k}h_{i,k} + \phi_i(y_i)\theta_i + \psi_i(y_i).\end{aligned} \tag{17}$$

Define the estimation error as $\varepsilon_i = x_i - \tilde{x}_i$, then by (17), after some mathematical derivations, one obtains

$$\dot{\varepsilon}_i = A_{i,0}\varepsilon_i - \dot{\Xi}_i\digamma_i - \dot{w}_a(t)\sum_{k=0}^{m_i} b_{i,k}h_{i,k}. \tag{18}$$

By (1), (16), it follows that

$$\begin{aligned}\dot{y}_i =& \varepsilon_{i,2} + \Xi_{i,2}\digamma_i + \Gamma_{i,2}\theta_i + \xi_{i,2} + b_{i,m_i}\eta_{i,(m_i,2)} + w_i^T\Theta_i \\ &+ \sum_{k=0}^{m_i} w_a b_{i,k}h_{i,(k,2)} + \phi_{i(1)}(y_i)\theta + \psi_{i(1)}(y_i)\end{aligned} \tag{19}$$

where $w_i = [0, \eta_{i,(m_i-1,2)}, \ldots, \eta_{i,(0,2)}]^T$, $\Theta_i = [b_{i,m_i}, b_{i,m_i-1}, \ldots, b_{i,0}]^T$, $\eta_{i,(k,2)}$, $\varepsilon_{i,2}$, $\Xi_{i,2}$, $\Gamma_{i,2}$, $\xi_{i,2}$, $h_{i,(k,2)}$ are all the second row of $\eta_{i,k}$, ε_i, Ξ_i, Γ_i, ξ_i, $h_{i,k}$, $k = 0, 1, \ldots, m_i$.

Remark 2. Due to the impact of attacks, only the false output $\check{x}_{i,1}$ can be used for observer design. In order to address these issues, with standard output feedback control approach (Zhou et al., 2011), the so-called state observer was constructed by introducing the information of compromised output in Liu et al. (2021), Wang et al. (2023).

$$\begin{aligned}\dot{\hat{x}}_i &= A_{i0}\hat{x}_i + D_i u_i - L_i\check{x}_{i,1} \\ \hat{y}_i &= C_i^T\hat{x}_i\end{aligned} \tag{20}$$

where $\hat{\tilde{x}}_i = [\hat{x}_{i,1}, \ldots, \hat{x}_{i,n}]^T$ is the estimate of x_i, $D_i = \begin{bmatrix}0, 0, \ldots, 1\end{bmatrix}^T$, $C_i^T = \begin{bmatrix}1, 0, \ldots, 0\end{bmatrix}$, $A_{i0} = A_i + L_iC_i^T$, $L_i = \begin{bmatrix}-l_{i,1}, -l_{i,2}, \ldots, -l_{i,n}\end{bmatrix}^T$ and $A_i = \begin{bmatrix} 0 & \\ \vdots & I_{n_i-1} \\ 0 & \ldots \quad 0 \end{bmatrix}$, L_i is the design matrix such that the matrix A_{i0} is a strict Hurwitz matrix. With these designed observer, the observer error equation is given as

$$\dot{e}_i = A_{i0}e_i + F(x_i) + L_i\omega_s(t, x_{i,1}). \tag{21}$$

Although the observer performs well in estimating the unknown states, however, in Liu et al. (2021), Wang et al. (2023), the compensation mechanism based on false output information is difficult to design, and the attack signal $\omega_s(t, x_1)$ is simply treated as $2e_i^TP_iL_i\omega_s(t, x_{i,1}) \leq \|e_i\|^2 + \|P_i\|^2\|L_i\|^2\bar{\omega}_1^2$, where the term $\omega_s(t, x_{i,1})$ related to attacks cannot be compensated, and the bounded results can be obtained only. When the attack occurs, the expected control performance cannot be achieved. Under the circumstance, it is still a challenge work to design a suitable state estimator. In this work, a new form of K-filters in (9)–(13) with the false output and a virtual estimate term in (16) are proposed such that the attack signals can be effectively compensated.

3.2. The design of the resilient adaptive controller

Now we replace (1) with a new system, whose states depend on filters (9)–(13), and thus are available for control design under deception attacks:

$$\begin{aligned}\dot{y}_i =& \varepsilon_{i,2} + \Xi_{i,2}\digamma_i + \Gamma_{i,2}\theta_i + \xi_{i,2} + b_{i,m_i}\eta_{i,(m_i,2)} + w_i^T\Theta_i \\ &+ \sum_{k=0}^{m_i} w_a b_{i,k}h_{i,(k,2)} + \phi_{i(1)}(y)\theta + \psi_{i(1)}(y) \\ \dot{\eta}_{i,(m_i,q)} =& \eta_{i,(m_i,q+1)} - k_{i,q}\eta_{i,(m_i,1)}, q = 1, \ldots, r_i - 1 \\ \dot{\eta}_{i,(m_i,r_i)} =& k_{i,r_i}\eta_{i,(m_i,1)} + q(u_i)\end{aligned} \tag{22}$$

where $\eta_{i,(m_i,q)}$, $q = 1, \ldots, r_i$ is the qth element of η_{i,m_i}. The following control design follows the backstepping design procedure through which the feedback control is recursively derived at the final step r_i. Nevertheless, different from the standard backstepping technique, a new coordinate transformation of the error is introduced:

$$\begin{aligned}z_{i,1} =& \sum_{j=1}^{N} a_{ij}(y_i - y_j) + b_i(y_i - y_d) \\ z_{i,q} =& \eta_{i,(m_i,q)} - \alpha_{i,(q-1)}, q = 2, \ldots, r_i.\end{aligned} \tag{23}$$

By defining $z_1 = [z_{1,1}, \ldots, z_{N,1}]^T$, one has

$$z_1 = (L + B)e \tag{24}$$

where $e = \underline{y} - \underline{y}_d$ with $\underline{y} = [y_1, \ldots, y_N] \in R^N$ and $\underline{y}_d = [y_d, \ldots, y_d] \in R^N$ denoting the actual tracking error between each subsystem's output y_i and y_d, $B = \text{diag}\{b_i\} \in R^{N\times N}$. $b_i = 1$ indicates that the desired trajectory is accessible directly by agent i; otherwise, $b_i = 0$. $\alpha_{i,(q-1)}$ is a virtual control signal to be specified later.

Remark 3. It should be pointed out that the error surface $z_{i,1}$ in (23) cannot be applied to the process of controller design, which is only appeared in the process of achieving the stabilization of the closed-loop system. This is a clearly differentiate between the general recursive design and the presented recursive design strategy for nonlinear MASs subject to deception attacks. Now, the backstepping design procedure is presented step by step.

***Step 1*:** The Lyapunov function V_1 is chosen as

$$V_1 = \frac{\lambda^2}{2\lambda_M} e^T(L+B)e + \sum_{i=1}^{N} \frac{1}{2\delta_{i,1}} \tilde{\gamma}_i^2 \tag{25}$$

where $\delta_i > 0$ is a design parameter, $\tilde{\gamma}_i = \gamma_i - \hat{\gamma}_i$, $\hat{\gamma}_i$ is the estimate of γ_i which will be defined later.

From (16)–(23), the derivative of V_1 is given as

$$\begin{aligned}\dot{V}_1 =& \frac{\lambda\dot{\lambda}}{\lambda_M} e^T(L+B)e + \frac{\lambda^2 z_{i,1}}{\lambda_M} \sum_{i=1}^{N}\Big[\varepsilon_{i,(a,2)} - \varepsilon_{i,(b,2)} - \varepsilon_{i,(c,2)} \\ &+ \Xi_{i,2}\Gamma_i + \Gamma_{i,2}\theta_i + \xi_{i,2} + b_{i,m_i} z_{i,2} + b_{i,m_i}\alpha_{i,1} \\ &+ w_i^T \Theta_i + \sum_{k=0}^{m_i} w_a b_{i,k} h_{i,(k,2)} + \phi_{i(1)}(y_i)\theta_i + \psi_{i(1)}(y_i) \\ &- \dot{y}_d\Big] - \sum_{i=1}^{N} \frac{1}{\delta_{i,1}} \tilde{\gamma}_i \dot{\hat{\gamma}}_i \end{aligned} \tag{26}$$

where $\varepsilon_{i,2} = \varepsilon_{i,(a,2)} - \varepsilon_{i,(b,2)} - \varepsilon_{i,(c,2)}$, $\varepsilon_{i,(a,2)}$, $\varepsilon_{i,(b,2)}$ and $\varepsilon_{i,(c,2)}$ are the second element of $\varepsilon_{i,a}$, $\varepsilon_{i,b}$ and $\varepsilon_{i,c}$. $\varepsilon_{i,a} = e^{A_{i,0}t}\varepsilon_{i,a}(0)$, $\varepsilon_{i,b} = \int_0^t e^{A_{i,0}(t-\tau)} \dot{\Xi}_i(\tau)\Gamma_i(\tau)d\tau$, $\varepsilon_{i,c} = \sum_{k=0}^{m_i} \int_0^t e^{A_{i,0}(t-\tau)} b_{i,k}\dot{w}_a(\tau) h_{i,k}(\tau)d\tau$. Then from (18), it is easy to obtain that $\varepsilon_i = \varepsilon_{i,a} - \varepsilon_{i,b} - \varepsilon_{i,c}$. Further, we have

$$\begin{aligned}|\varepsilon_{i,(b,2)}| + |\varepsilon_{i,(c,2)}| \le& \int_0^t \|e_{n_i,2}^T e^{A_{i,0}(t-\tau)}\| \|\dot{\Xi}_i(\tau)\| \|\Gamma_i(\tau)\| d\tau \\ &+ \sum_{k=0}^{m_i} \int_0^t \|e_{n_i,2}^T e^{A_{i,0}(t-\tau)}\| \|b_{i,k}\dot{w}_a(\tau)\| \|h_{i,k}(\tau)\| d\tau \\ &\le \gamma_{i,s}\varrho_i \end{aligned} \tag{27}$$

where $\gamma_{i,s} = \max\{\gamma_{i,b}, \gamma_{i,c}\}$, $\gamma_{i,b} = \sup\|\dot{\Xi}_i(\tau)\|$, $\gamma_{i,c} = \max\{|b_{i,k}|\bar{w}_a, k = 0, \ldots, m_i\}$. ϱ_i is the auxiliary variable, which, is designed as $\dot{\varrho}_i = -r_{i,1}\varrho_i + r_{i,2}(\|\Gamma_i(\tau)\| + \sum_{k=0}^{m_i}\|h_{i,k}(\tau)\|)$ with $\varrho_i(0) \ge 0$, $r_{i,1}$ and $r_{i,2}$ are the positive constants.

According to Lemma 3, as for $\phi_{i(1)}(y_i)$, there exist $h_\phi(\cdot)$ and $d_{i,1}(\cdot)$ such that $\|\phi_{i(1)}(y_i)\| = \|\phi_{i(1)}(\lambda\check{x}_{i,1})\| \le h_\phi(\lambda) d_{i,1}(\check{x}_{i,1}) \le \lambda_\phi d_{i,1}(\check{x}_{i,1})$, where λ_ϕ is the unknown constant bound of $h_\phi(\lambda)$ and $d_{i,1}(\check{x}_{i,1})$ is a known smooth function. Therefore, by using Lemma 3, we can get

$$\frac{\lambda^2 z_{i,1}}{\lambda_M}\phi_{i(1)}(y_i)\theta_i \le \gamma_{i,e}(\sigma + v_{i,e,1}\check{z}_{i,1}) \tag{28}$$

where $\check{z}_{i,1} = \sum_{j=1}^{N} a_{ij}(\check{y}_i - \check{y}_j) + b_i(\check{y}_i - \check{y}_d)$, $\gamma_{i,e} = \lambda_M^2\lambda_\phi\|\theta_i\|$, $v_{i,e,1} = \frac{\check{z}_{i,1} d_{i,1}^2(\check{z}_{i,1})}{\sqrt{[\check{z}_{i,1} d_{i,1}(\check{z}_{i,1})]^2 + \sigma^2}}$. Similarly, using Young's inequality and Lemma 4, we have

$$\begin{aligned}\dot{V}_1 \le& \sum_{i=1}^{N}\Big[\check{z}_{i,1}\gamma_i v_{i,1} + \frac{1}{4}z_{i,2}^2 + \frac{\lambda^2 b_{i,m_i}}{\lambda_M} z_{i,1}\alpha_{i,1} + \gamma_i\sigma_{i,s} \\ &+ \frac{1}{4}\varepsilon_{i,a}^T\varepsilon_{i,a} - \frac{1}{\delta_{i,1}}\tilde{\gamma}_i\dot{\hat{\gamma}}_i\Big] \end{aligned} \tag{29}$$

where $\sigma_{i,s} = (9 + m_i)\sigma$, $\sigma(t)$ is a positive uniformly continuous and bounded function for $t \ge 0$, and the inequality $\lim_{t\to\infty}\int_0^t \sigma(\tau)d\tau \le \bar{\sigma} < +\infty$ holds, where $\bar{\sigma} > 0$ is a constant. $\gamma_i = \max\{\gamma_{i,a}, \gamma_{i,b}, \gamma_{i,c}, \gamma_{i,d}, \gamma_{i,e}, \gamma_{i,f}, \gamma_{i,g}, \gamma_{i,h}, \gamma_{i,p}, \gamma_{i,w}, \gamma_{i,l}\}$, $v_{i,1} = v_{i,a,1} + v_{i,b,1} + v_{i,c,1} + v_{i,d,1} + v_{i,e,1} + v_{i,f,1} + v_{i,g,1} + v_{i,h,1} + v_{i,p,1} + v_{i,w,1} + v_{i,l,1} + v_{i,u,1}$, $\gamma_{i,a} = \lambda_M^4$, $\gamma_{i,b} = \lambda_M^2\|\theta_i\|$, $\gamma_{i,c}$ is the upper bound of $\|\lambda_M^2\Xi_{i,2}\|$, $\gamma_{i,d} = \lambda_M^2$, $\gamma_{i,f} = \lambda_M^2\lambda_\psi$, $\gamma_{i,g} = \lambda_M^2\gamma_{i,s}$, $\gamma_{i,h} = \lambda_M^2\dot{\lambda}/\underline{\sigma}(L+B)$ where $\underline{\sigma}(L+B)$ is the minimum singular value of $L+B$, $\gamma_{i,p} = b_{i,m_i}^2\lambda_M^4$, $\gamma_{i,w} = \|\lambda_M^2\Theta_i\|$, $\gamma_{i,l} = [\gamma_{i,l0}, \ldots, \gamma_{i,lm_i}]^T$ with $\gamma_{i,lj} = \lambda_M^2\bar{w}_a|b_{i,j}|$, $j = 0, \ldots, m_i$, $\gamma_{i,u} = \lambda_M^2 Y_d$. $v_{i,a,1} = \check{z}_{i,1}$, $v_{i,b,1} = \frac{\check{z}_{i,1}\Gamma_{i,2}^T\Gamma_{i,2}}{\sqrt{\check{z}_{i,1}^2\Gamma_{i,2}^T\Gamma_{i,2}+\sigma^2}}$, $v_{i,c,1} = \frac{\check{z}_{i,1}F_i^T F_i}{\sqrt{\check{z}_{i,1}^2 F_i^T F_i+\sigma^2}}$, $v_{i,d,1} = \frac{\check{z}_{i,1}\xi_{i,2}^2}{\sqrt{(\check{z}_{i,1}\xi_{i,2})^2+\sigma^2}}$, $v_{i,f,1} = \frac{\check{z}_{i,1}q_{i,1}^2(\check{z}_{i,1})}{\sqrt{[\check{z}_{i,1}q_{i,1}(\check{z}_{i,1})]^2+\sigma^2}}$, $v_{i,g,1} = \frac{\varrho_i^2\check{z}_{i,1}}{\sqrt{(\varrho_i\check{z}_{i,1})^2+\sigma^2}}$, $v_{i,h,1} = \check{z}_{i,1}$, $v_{i,p,1} = \check{z}_{i,1}$, $v_{i,w,1} = \frac{\check{z}_{i,1}w_i^T w_i}{\sqrt{\check{z}_{i,1}^2 w_i^T w_i+\sigma^2}}$, $v_{i,l,1} = [\frac{\check{z}_{i,1}h_{i,(0,2)}^2}{\sqrt{(\check{z}_{i,1}h_{i,(0,2)})^2+\sigma^2}}, \ldots, \frac{\check{z}_{i,1}h_{i,(m_i,2)}^2}{\sqrt{(\check{z}_{i,1}h_{i,(m_i,2)})^2+\sigma^2}}]^T$, $v_{i,u,1} = \frac{\check{z}_{i,1}}{\sqrt{\check{z}_{i,1}^2+\sigma^2}}$.

Design the virtual control $\alpha_{i,1}$ as

$$\alpha_{i,1} = N_i(\kappa_i)\Phi_i, \dot{\kappa}_i = \check{z}_{i,1}\Phi_i, \Phi_i = \bar{s}_{i,1}\check{z}_{i,1} + \hat{\gamma}_i v_{i,1}. \tag{30}$$

By substituting (35) into (26), one can obtain

$$\begin{aligned}\dot{V}_1 \le& \sum_{i=1}^{N}\Big[-s_{i,1}z_{i,1}^2 + \tilde{\gamma}_i\check{z}_{i,1}v_{i,1} + (\beta_i(t)N_i + 1)\dot{\kappa}_i - \frac{1}{\delta_{i,1}}\tilde{\gamma}_i\dot{\hat{\gamma}}_i \\ &+ \frac{1}{4}z_{i,2}^2 + \gamma_i\sigma_{i,s} + \frac{1}{4}\varepsilon_{i,a}^T\varepsilon_{i,a}\Big] \end{aligned} \tag{31}$$

where $s_{i,1} = \bar{s}_{i,1}\lambda_m^{-2}$, $\beta_i(t) = \lambda_M^{-1}\lambda^3 b_{i,m_i}$.

***Step 2*:** Differentiating $z_{i,2}$, we obtain

$$\begin{aligned}\dot{z}_{i,2} =& z_{i,3} + \alpha_{i,2} - k_{i,2}\eta_{i,(m_i,1)} - \frac{1}{\lambda}\frac{\partial\alpha_{i,1}}{\partial\check{x}_{i,1}}(\varepsilon_{i,(a,2)} - \varepsilon_{i,(b,2)} \\ &- \varepsilon_{i,(c,2)} + \Xi_{i,2}\Gamma_i + \Gamma_{i,2}\theta_i + \xi_{i,2} + b_{i,m_i}\eta_{i,(m_i,2)} \\ &+ w_i^T\Theta_i + \phi_{i(1)}(y)\theta_i + \psi_{i(1)}(y) + \sum_{k=0}^{m_i} w_a b_{i,k} h_{i,(k,2)}) \\ &- \frac{1}{\lambda}\frac{\partial\alpha_{i,1}}{\partial\check{x}_{j,1}}(\phi_{j(1)}\theta_j + \psi_{j(1)}) - \frac{1}{\lambda}\frac{\partial\alpha_{i,1}}{\partial\check{y}_d}\dot{y}_d \\ &+ \frac{\dot{\lambda}}{\lambda}(\frac{\partial\alpha_{i,1}}{\partial\check{x}_{i,1}}\check{x}_{i,1} + \frac{\partial\alpha_{i,1}}{\partial\check{x}_{j,1}}\check{x}_{j,1} + \frac{\partial\alpha_{i,1}}{\partial\check{y}_d}\check{y}_d) \\ &- \Lambda_{i,2} \end{aligned} \tag{32}$$

where $\Lambda_{i,2} = \frac{\partial\alpha_{i,1}}{\partial\hat{\gamma}_i}\dot{\hat{\gamma}}_i + \frac{\partial\alpha_{i,1}}{\partial\sigma}\dot{\sigma} + \frac{\partial\alpha_{i,1}}{\partial\Gamma_i}(A_{i,0}\Gamma_i + \phi_i(\check{x}_{i,1})) + \frac{\partial\alpha_{i,1}}{\partial\xi_i}(A_{i,0}\xi_i + \psi_i(\check{x}_{i,1})) + \frac{\partial\alpha_{i,1}}{\partial F_i}(A_{i,0}F_i + H_{i,0}\check{x}_{i,1}) + \sum_{k=1}^{m_i}\frac{\partial\alpha_{i,1}}{\partial v_{i,k}}(-k_{i,k}v_{i,1} + v_{i,k+1}) + \frac{\partial\alpha_{i,1}}{\partial\varrho_i}\dot{\varrho}_i + \frac{\partial\alpha_{i,1}}{\partial\kappa_i}\dot{\kappa}_i + \sum_{k=0}^{m_i}\frac{\partial\alpha_{i,1}}{\partial h_{i,k,2}}\dot{h}_{i,k,2} + \sum_{j=1}^{N} a_{ij}\frac{\partial\alpha_{i,1}}{\partial x_{j,1}}\check{x}_{j,2}$.

Choose the second Lyapunov function candidate as

$$V_2 = V_1 + \sum_{i=1}^{N}\Big[\frac{1}{2}z_{i,2}^2 + \frac{1}{2\delta_{i,2}}\tilde{\rho}_i^2\Big] \tag{33}$$

where $\delta_{i,2} > 0$ is a design parameter, $\tilde{\rho}_i = \rho_i - \hat{\rho}_i$, $\hat{\rho}_i$ is the estimate of ρ_i.

In light of (32), the derivative of V_2 is computed as

$$\begin{aligned}\dot{V}_2 \le& \dot{V}_1 + \sum_{i=1}^{N}\Big[z_{i,2}\rho_i v_{i,2} + z_{i,2}(z_{i,3} + \alpha_{i,2}) + \rho_i\sigma_{i,p} \\ &+ \frac{1}{4}\varepsilon_{i,a}^T\varepsilon_{i,a} - \frac{1}{\delta_{i,2}}\tilde{\rho}_i\dot{\hat{\rho}}_i\Big] \end{aligned} \tag{34}$$

where $\rho_i = \max\{\rho_{i,a}, \rho_{i,b}, \rho_{i,c}, \rho_{i,d}, \rho_{i,a}, \rho_{i,f}, \rho_{i,g}, \rho_{i,h}, \rho_{i,p}, \rho_{i,w}, \rho_{i,l}, \rho_{i,o}, \rho_{i,u}\}$, $\rho_{i,a} = \lambda_m^{-2}$, $\rho_{i,b} = \lambda_m^{-1}\|\theta_i\|$, $\rho_{i,c}$ is the upper bound of $\|\lambda_m^{-1}\Xi_{i,2}\|$, $\rho_{i,d} = \lambda_m^{-1}$, $\rho_{i,e} = \lambda_m^{-1}\lambda_\phi\|\theta\|$, $\rho_{i,f} = \lambda_m^{-1}\lambda_\psi$, $\rho_{i,g} = \lambda_m^{-1}\gamma_{i,s}$, $\rho_{i,h} = \lambda_m^{-1}\bar{\lambda}$, $\rho_{i,p} = \lambda_m^{-1}b_{i,m_i}$, $\rho_{i,w} = \|\lambda_m^{-1}\Theta_i\|$, $\rho_{i,l} = [\rho_{i,l0}, \ldots, \rho_{i,lm_i}]$ with $\rho_{i,lk} = \lambda_m^{-1}\bar{w}_a|b_{i,k}|$, $k = 0, \ldots, m_i$, $\rho_{i,o} = \lambda_m^{-1}\lambda_\phi\theta_{jmax}$ with $\theta_{jmax} = \max\{\|\theta_j\|, j \in N_i\}$, $\rho_{i,u} = \lambda_m^{-1}Y_d$.

For $q = 2, \ldots, r$, $v_{i,q} = v_{i,a,q} + v_{i,b,q} + v_{i,c,q} + v_{i,d,q} + v_{i,e,q} + v_{i,f,q} + v_{i,g,q} + v_{i,h,q} + v_{i,p,q} + v_{i,w,q} + v_{i,l,q} + v_{i,o,q} + v_{i,u,q}$, $v_{i,a,q} = z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2$, $v_{i,b,q} = z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 \Gamma_{i,2}^T \Gamma_{i,2}[(z_{i,q}\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}}\Gamma_{i,2})^2 + \sigma^2]^{-1/2}$, $v_{i,c,q} = z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 F_i^T F_i[z_{i,q}^2(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 F_i^T F_i + \sigma^2]^{-1/2}$, $v_{i,d,q} = z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 \xi_{i,1}^2[(z_{i,q}\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}}\xi_{i,1})^2 + \sigma^2]^{-1/2}$, $v_{i,e,q} = z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 d_{i,1}^2(\check{x}_{i,1})[[z_{i,q}\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}}d_1(\check{x}_{i,1})]^2 + \sigma^2]^{-1/2}$, $v_{i,f,q} = z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 q_{i,1}^2(\check{x}_{i,1})[[z_{i,q}\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}}q_{i,1}(\check{x}_{i,1})]^2 + \sigma^2]^{-1/2} + \sum_{j\in N_i}\sum_{k=1}^{q-1} z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{j,k}})^2 q_{j,k}^2[(z_{i,q}\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{j,k}}q_{j,k})^2 + \sigma^2]^{-1/2}$, $v_{i,g,q} = z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 \varrho_i^2[(z_{i,q}\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}}\varrho_i)^2 + \sigma^2]^{-1/2}$, $v_{i,h,q} = z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}}\check{x}_{i,1} + \sum_{k=1}^{q-1}\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,k}}\check{x}_{j,k} + \frac{\partial \alpha_{i,q-1}}{\partial \check{y}_d}\check{y}_d)^2[z_{i,q}^2(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}}\check{x}_{i,1} + \sum_{k=1}^{q-1}\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,k}}\check{x}_{j,k} + \frac{\partial \alpha_{i,q-1}}{\partial \check{y}_d}\check{y}_d)^2 + \sigma^2]^{-1/2}$, $v_{i,p,q} = z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 \eta_{i,m_i,2}^2[z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 \eta_{i,m_i,2}^2 + \sigma^2]^{-1/2}$, $v_{i,w,q} = z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 w_i^T w_i[z_{i,q}^2(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 w_i^T w_i + \sigma^2]^{-1/2}$, $v_{i,l,q} = [z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 h_{i,(0,2)}^2[z_{i,q}^2(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 h_{i,(0,2)}^2 + \sigma^2]^{-1/2}, \ldots, z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 h_{i,(m_i,2)}^2[z_{i,q}^2(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{i,1}})^2 h_{i,(m_i,2)}^2 + \sigma^2]^{-1/2}]^T$, $v_{i,o,q} = \sum_{j\in N_i}\sum_{k=1}^{q-1} z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{j,k}})^2 d_{j,k}^2[(z_{i,q}\frac{\partial \alpha_{i,q-1}}{\partial \check{x}_{j,k}}d_{j,k})^2 + \sigma^2]^{-1/2}$, $v_{i,u,q} = z_{i,q}(\frac{\partial \alpha_{i,q-1}}{\partial \check{y}_d})^2[(z_{i,q}\frac{\partial \alpha_{i,q-1}}{\partial \check{y}_d})^2 + \sigma^2]^{-1/2}$.

The virtual controller $\alpha_{i,2}$ is constructed as

$$\alpha_{i,2} = -(s_{i,2} + \frac{1}{4})z_{i,2} - \hat{\rho}_i v_{i,2} + \Lambda_{i,2} + k_{i,2}\eta_{i,(m_i,1)} \tag{35}$$

where $s_{i,2}$ is a positive design parameter selected freely by the designer, let the tuning function $\varsigma_{i,2}$ as $\varsigma_{i,2} = \delta_{i,2} z_{i,2} v_{i,2}$, which yields

$$\begin{aligned}\dot{V}_2 \le & \sum_{i=1}^{N}\Big[\sum_{l=1}^{2} -s_{i,l}z_{i,l}^2 + \tilde{\gamma}_i\check{z}_{i,1}v_{i,1} + (\beta_i(t)N_i + 1)\dot{\kappa}_i \\ & - \frac{1}{\delta_{i,1}}\tilde{\gamma}_i\dot{\hat{\gamma}}_i + \gamma_i\sigma_{i,s} + \frac{1}{2}\varepsilon_{i,a}^T\varepsilon_{i,a} + \frac{1}{\delta_{i,2}}\tilde{\rho}_i(\varsigma_{i,2} - \dot{\hat{\rho}}_i) \\ & + z_{i,2}z_{i,3} + \rho_i\sigma_{i,p}\end{aligned} \tag{36}$$

where $\sigma_{i,p} = (12 + m_i + M_i(q-1))\sigma$, M_i is the num of neighbors for node i.

Step q ($3 \le q \le r_i - 1$): By following the ideas in the above two steps, the virtual control laws are shown in Table 1, and with the help of the Lyapunov function $V_q = V_{q-1} + \sum_{i=1}^{N}\frac{1}{2}z_{i,q}^2$, one has

$$\begin{aligned}\dot{V}_q \le & \sum_{i=1}^{N}\Big[\sum_{l=1}^{q} -s_{i,l}z_{i,l}^2 + \tilde{\gamma}_i\check{z}_{i,1}v_{i,1} + (\beta_i(t)N_i + 1)\dot{\kappa}_i \\ & - \frac{1}{\delta_{i,1}}\tilde{\gamma}_i\dot{\hat{\gamma}}_i + \gamma_i\sigma_{i,s} + \frac{q}{4}\varepsilon_{i,a}^T\varepsilon_{i,a} + z_{i,q}z_{i,q+1} \\ & - \sum_{l=3}^{q} z_{i,l}\frac{\partial \alpha_{i,l-1}}{\partial \hat{\rho}_i}(\dot{\hat{\rho}}_i - \varsigma_{i,q}) + \frac{1}{\delta_{i,2}}\tilde{\rho}_i(\varsigma_{i,q} - \dot{\hat{\rho}}_i) \\ & - \frac{q}{4}\varepsilon_{i,a}^T\varepsilon_{i,a} + (q-1)\rho_i\sigma_{i,p}.\end{aligned} \tag{37}$$

Step r_i: The final controller and adaptive laws are recursively derived at the final step r_i, as summarized in Table 1. From (23) and using (T1.3), the derivative of z_{i,r_i} is described as $\dot{z}_{i,r_i} = -s_{i,r_i}z_{i,r_i} - z_{i,r_i-1} + q(u_i) + \alpha_{i,r_i}$. Consider the Lyapunov function $V_{r_i} = V_{r_i-1} + \sum_{i=1}^{N}\frac{1}{2}z_{i,r_i}^2$, from (37) and (T1.5), which yields

$$\begin{aligned}\dot{V}_{r_i} \le & \sum_{i=1}^{N}\Big[\sum_{l=1}^{r_i} -s_{i,l}z_{i,l}^2 + (\beta_i(t)N_i + 1)\dot{\kappa}_i + \gamma_i\sigma_{i,s} \\ & + (r_i - 1)\rho_i\sigma_{i,p} + z_{i,r_i}\alpha_{i,r_i} + z_{i,r_i}(u_i + d_i) \\ & - \frac{r_i}{4}\varepsilon_{i,a}^T\varepsilon_{i,a}\Big].\end{aligned} \tag{38}$$

Noting (T1.4) that $z_{i,r_i}u_i < 0$ and considering Lemma 4, it yields

$$\begin{aligned}z_{i,r_i}(u_i + d_i) &\le z_{i,r_i}u_i + \delta|z_{i,r_i}u_i| + u_{i,\min}|z_{i,r_i}| \\ &\le (1-\delta)z_{i,r_i}u_i + u_{i,\min}(\frac{z_{i,r_i}^2}{\sqrt{z_{i,r_i}^2 + \sigma^2}} + \sigma) \\ &\le -\frac{z_{i,r_i}^2\alpha_{i,r_i}^2}{\sqrt{z_{i,r_i}^2\alpha_{i,r_i}^2 + \sigma^2}} + u_{i,\min}\sigma \\ &\le \sigma - z_{i,r_i}\alpha_{i,r_i} + u_{i,\min}\sigma.\end{aligned} \tag{39}$$

Then, we get

$$\begin{aligned}\dot{V}_r \le & \sum_{i=1}^{N}\Big[\sum_{l=1}^{r_i} -s_{i,l}z_{i,l}^2 + (\beta_i(t)N_i + 1)\dot{\kappa}_i + \gamma_i\sigma_{i,s} \\ & + (r_i - 1)\rho_i\sigma_{i,p} + (u_{i,\min} + 1)\sigma - \frac{r_i}{4}\varepsilon_{i,a}^T\varepsilon_{i,a}\Big].\end{aligned} \tag{40}$$

3.3. System stability

The main results of our distributed adaptive control design scheme can be formally stated in the following theorem.

Theorem 1. *Consider a class of nonlinear MASs consisting of N agents (1) suffering from unknown sensor and actuator attacks. Suppose that Assumptions 1–4 hold, if the filters (9)–(13), the resilient adaptive controller (T1.4), the adaptive update laws (T1.5) are applied, then all the signals of the closed-loop systems are globally uniformly bounded and all the N subsystems' outputs can asymptotically track a desired trajectory $y_d(t)$, i.e., $\lim_{t\to\infty} e_i(t) = 0$ for $i = 1, \ldots, N$.*

Proof. In order to analyze the stability of the overall system, the Lyapunov function V is constructed as

$$V = V_{r_i} + \frac{r_i}{4}\varepsilon_{i,a}P_i\varepsilon_{i,a} \tag{41}$$

where $P_i > 0$ is a positive definite matrix such that $P_iA_{i,0} + A_{i,0}^TP_i = -I_{n_i}$.

From (40), we obtain

$$\begin{aligned}\dot{V} \le & \sum_{i=1}^{N}\Big[\sum_{l=1}^{r_i} -s_{i,l}z_{i,l}^2 + (\beta_i(t)N_i + 1)\dot{\kappa}_i + \gamma_i\sigma_{i,s} + (r_i - 1)\rho_i\sigma_{i,p} \\ & + (u_{i,\min} + 1)\sigma\Big] \\ \le & -\sum_{l=1}^{r_i} s_l\|z_l\|^2 + \sum_{i=1}^{N}\Big[(\beta_i(t)N_i + 1)\dot{\kappa}_i + \gamma_i\sigma_{i,s} \\ & + (r_i - 1)\rho_i\sigma_{i,p} + (u_{i,\min} + 1)\sigma\Big]\end{aligned} \tag{42}$$

where $s_l = \min\{s_{i,l}\}$.

By integrating the differential inequality (42) on $[0, t]$, we have

$$\begin{aligned}V(t) \le & V(0) + \sum_{i=1}^{N}\int_0^t(\beta_i(t)N_i(\kappa_i) + 1)\dot{\kappa}_i ds \\ & - \sum_{l=1}^{r_i} s_l\int_0^t\|z_l\|^2 ds + \Upsilon \\ \le & V(0) + \sum_{i=1}^{N}\int_0^t(\beta_i(t)N_i(\kappa_i) + 1)\dot{\kappa}_i ds + \Upsilon\end{aligned} \tag{43}$$

where $\Upsilon = \sum_{i=1}^{N}((9+m)(r_i-1)\rho + (7+m)\gamma + u_{i,\min} + 1)\bar{\sigma}$ is a suitable positive constant. By Lemma 2, we know $V(t)$, $\kappa_i(t)$ and

Table 1
The design of controllers and adaptive laws.

Step q: $(q = 3)$	
Virtual control Law: $\alpha_{i,3} = -s_{i,3}z_{i,3} - z_{i,2} + k_{i,3}\eta_{i,m_i,1} + \Lambda_{i,3} + \frac{\partial\alpha_{i,2}}{\partial\hat{\rho}}\delta_{i,2}\varsigma_{i,3}$ with $\varsigma_{i,3} = \varsigma_{i,2} + \delta_{i,2}z_{i,3}v_{i,3}$, $\Lambda_{i,3} = \frac{\partial\alpha_{i,2}}{\partial\hat{\gamma}_i}\dot{\hat{\gamma}}_i + \frac{\partial\alpha_{i,2}}{\partial\sigma}\dot{\sigma} + \frac{\partial\alpha_{i,2}}{\partial\Gamma_i}(A_{i,0}\Gamma_i + \phi_i(\check{x}_{i,1})) + \frac{\partial\alpha_{i,2}}{\partial\xi_i}(A_{i,0}\xi_i + \psi_i(\check{x}_{i,1})) + \frac{\partial\alpha_{i,2}}{\partial F_i}(A_{i,0}F_i + H_{i,0}\check{x}_{i,1})$ $+ \sum_{k=1}^{m_i+1}\frac{\partial\alpha_{i,2}}{\partial v_{i,k}}(-k_{i,k}v_{i,1} + v_{i,k+1}) + \frac{\partial\alpha_{i,2}}{\partial\varrho_i}\dot{\varrho}_i + \frac{\partial\alpha_{i,2}}{\partial\kappa_i}\dot{\kappa}_i + \sum_{k=0}^{m_i}\frac{\partial\alpha_{i,2}}{\partial\hat{h}_{i,k,2}}\dot{\hat{h}}_{i,k,2} + \sum_{j\in N_i}\sum_{k=1}^{2}\frac{\partial\alpha_{i,2}}{\partial x_{j,k}}\check{x}_{j,k+1}$.	(T1.1)
Step q: $(q = 4, \ldots, r-1)$ Virtual control Laws: $\alpha_{i,q} = -s_{i,q}z_{i,q} - z_{i,q-1} + k_{i,q}\eta_{i,m_i,1} + \Lambda_{i,q} + \frac{\partial\alpha_{i,q-1}}{\partial\hat{\rho}_i}\varsigma_{i,q} + \delta_{i,2}(\sum_{k=3}^{q-1}z_{i,k}\frac{\partial\alpha_{i,k-1}}{\partial\hat{\rho}_i}z_{i,q}v_{i,q})$ with $\varsigma_{i,q} = \varsigma_{i,q-1} + \delta_{i,2}z_{i,q}^2v_{i,q}$, $\Lambda_{i,q} = \frac{\partial\alpha_{i,q-1}}{\partial\hat{\gamma}_i}\dot{\hat{\gamma}}_i + \frac{\partial\alpha_{i,q-1}}{\partial\sigma}\dot{\sigma} + \frac{\partial\alpha_{i,q-1}}{\partial\Gamma_i}(A_{i,0}\Gamma_i + \phi_i(\check{x}_{i,1})) + \frac{\partial\alpha_{i,q-1}}{\partial\xi_i}(A_{i,0}\xi_i + \psi_i(\check{x}_{i,1})) + \frac{\partial\alpha_{i,q-1}}{\partial F_i}(A_{i,0}F_i + H_{i,0}\check{x}_{i,1})$ $+ \sum_{k=1}^{m_i+q-2}\frac{\partial\alpha_{i,q-1}}{\partial v_{i,k}}(-k_{i,k}v_{i,1} + v_{i,k+1}) + \frac{\partial\alpha_{i,q-1}}{\partial\varrho_i}\dot{\varrho}_i + \frac{\partial\alpha_{i,q-1}}{\partial\kappa_i}\dot{\kappa}_i + \sum_{k=0}^{m_i}\frac{\partial\alpha_{i,q-1}}{\partial\hat{h}_{i,k,2}}\dot{\hat{h}}_{i,k,2} + \sum_{j\in N_i}\sum_{k=1}^{q-1}\frac{\partial\alpha_{i,q-1}}{\partial x_{j,k}}\check{x}_{j,k+1}$, $q = 4, \ldots, r$.	(T1.2)
Step r_i: Auxiliary variable: $\alpha_{i,r_i} = s_{i,r_i}z_{i,r_i} + z_{i,r_i-1} + \eta_{i,m_i,r_i+1} - k_{i,r_i}\eta_{i,m_i,1} - \Lambda_{i,r_i} - \sum_{l=3}^{r_i-1}z_{i,l}\frac{\partial\alpha_{i,l-1}}{\partial\hat{\rho}_i}\delta_{i,2}z_{i,r_i}^2v_{i,r_i} - \frac{\partial\alpha_{i,r_i-1}}{\partial\hat{\rho}_i}\varsigma_{i,r_i}$.	(T1.3)
Final control law: $u_i = -\frac{z_{i,r_i}\alpha_{i,r_i}^2}{(1-\delta)\sqrt{z_{i,r_i}^2\alpha_{i,r_i}^2+\sigma^2}} - \frac{u_{i,\min}z_{i,r_i}}{(1-\delta)\sqrt{z_{i,r_i}^2+\sigma^2}}$.	(T1.4)
Adaptive laws: $\dot{\hat{\gamma}}_i = \delta_{i,1}\check{z}_{i,1}v_{i,1}$, $\dot{\hat{\rho}}_i = \varsigma_{i,r_i}$, $\varsigma_{i,r_i} = \varsigma_{i,r_i-1} + \delta_{i,2}z_{i,r_i}^2v_{i,r_i}$.	(T1.5)

$\sum_{i=1}^{N}\int_0^t(\beta_iN_i(\kappa_i)+1)\dot{\kappa}_ids$ are bounded. According to the definition of $V(t)$, we can obtain that z_q, $q = 1, 2, \ldots, r_i$, $\tilde{\gamma}_i$, $\tilde{\rho}_i$, $\varepsilon_{i,a}$ are bounded, and $z_{i,q} \in \mathscr{L}_2$ for $i = 1, \ldots, N$, which further implies y_i, $\hat{\gamma}_i$, $\hat{\rho}_i$ are bounded. It follows from (9)–(11), it is clear that F_i, Γ_i and ξ_i are bounded, which leads to ϱ_i in (27) is bounded. According to the boundedness of y_i, and the smoothness of $\phi_{i,j}(y_i)$, $\psi_{i,j}(y_i)$ and $\Delta(y_i)$, Assumption 4, by following a recursive analysis similar to Chapter 8 in Krstic et al. (1995), one can get $v_{i,k}$, $k = 1, \ldots, m_i$ are bounded. From $\eta_{i,k} = A_{i,0}^kv_i$, it follows that $\eta_{i,k,l} = [*, \ldots, *][v_{i,1}, \ldots, v_{i,k+l}]^T$, which indicates that $v_{i,m_i+1} = \eta_{i,m_i,1} - [*, \ldots, *, 1][v_{i,1}, \ldots, v_{i,m_i}]^T$ is bounded. Using the same analytical method as above, it can be obtained that v_i is bounded, and then $\hat{x}_i$ and ε_i are bounded. Recalling (T1.1) - (T1.5), it can be shown that controller u_i is bounded. Thus, we prove that all the signals of the closed-loop system are globally uniformly bounded. By applying the Barbalat's lemma, it further obtains that $\lim_{t\to\infty}z_{i,q}(t) = 0$ for $i = 1, \ldots, N$ and $q = 1, \ldots, r_i$. From (23) and $(L + B)$ being positive definite, it follows that all the N subsystems' outputs can asymptotically track a desired trajectory y_d under deception attacks, i.e., $\lim_{t\to\infty}e_i(t) = 0$ for $i = 1, \ldots, N$.

Remark 4. Inspired by Yu and Lin (2016), in this paper, by proposing a novel transformation of the final controller u_i in (T1.4) rather than $-\alpha_{i,r_i}$ in (T1.3), where α_{i,r_i} is obtained via the backstepping design approach, the effect of the nonlinearity exists in the quantizer on the system stability can be effectively eliminated and the asymptotic stability under deception attacks is achieved.

4. Simulation studies

In this section, the stabilization problem of three single-link robots under sensor and actuator attacks is considered, which can be described by Huang and Chen (2004):

$$M_i\ddot{\vartheta}_i + \frac{1}{2}m_ig_il_i\sin(\vartheta_i) = u_i, y_i = \vartheta_i. \tag{44}$$

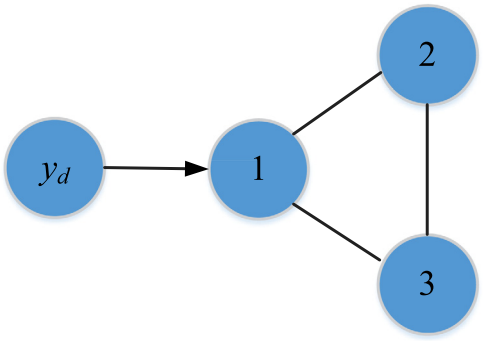


Fig. 2. Communication topology for a group of 3 agents.

In the simulation, we consider the hysteretic quantizer in (2). Let $x_{i,1} = \vartheta_i$, $x_{i,2} = \dot{\vartheta}_i$, then the system model can be rewritten as the output-feedback form:

$$\begin{aligned}\dot{x}_{i,1} &= x_{i,2}\\ \dot{x}_{i,2} &= \theta_i\phi_{i,2}(y_i) + b_{i,0}[q(u_i) + \rho_{i,a}(t, x_{i,1})]\\ y_i &= x_{i,1}\end{aligned} \tag{45}$$

where the unknown parameters are defined as $\theta_i = -\frac{1}{2M_i}m_ig_il_i$, $b_{i,0} = \frac{1}{M_i} > 0$. $M_i = 0.5$, $g_i = 9.8$, $m_i = 1$, $l_i = 1$, $\phi_{i,2}(y_i) = \sin(y_i)$. The physical significance for related parameters can refer to Huang and Chen (2004). In simulation, the initial states are selected as $x_{i,1}(0) = 0.1$, $x_{i,2}(0) = 0$. The attacks are chosen as $\zeta(t) = 1 + 0.25\cos(t)$, $\rho_{i,a}(t, x_1) = x_{i,1}\sin(1.5t)$. Quantizer parameters are chosen as $u_{i,\min} = 0.02$, $\mu = 0.01$. The design parameters are chosen as $\bar{s}_{i,1} = 1$, $s_{i,2} = 50$, $k_{i,1} = 0.1$, $k_{i,2} = 1$, $r_{i,1} = 0.1$, $r_{i,2} = 1$, $\delta_{i,1} = 0.001$, $\delta_{i,2} = 0.1$ for $1 \le i \le 3$.

The communication topology for 3 subsystems is given in Fig. 2. Simulation results are displayed in Figs. 3–6, respectively. Fig. 3 shows the tracking performance of all the subsystems' outputs $y_i(t) = x_{i,1}(t)$ for $1 \le i \le 3$ with comparison to

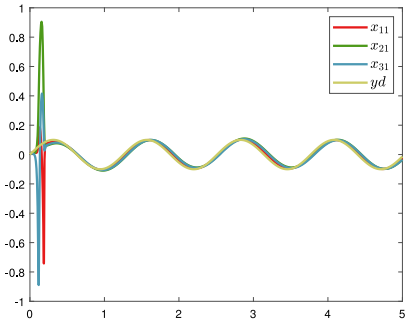


Fig. 3. Tracking performance of the outputs $y_i(t)$ for $1 \leq i \leq 3$ to the desired trajectory y_d.

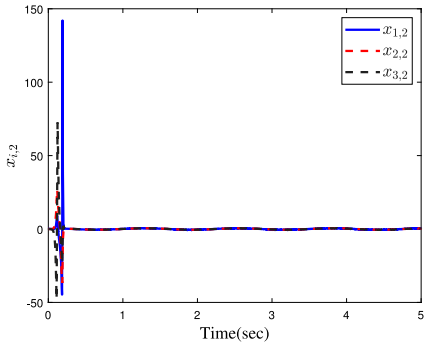


Fig. 4. The states $x_{i,2}(t)$ for $1 \leq i \leq 3$.

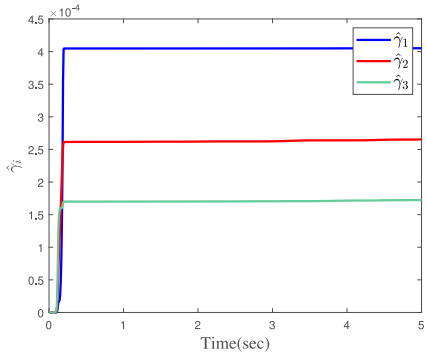


Fig. 5. Parameter estimates $\hat{\gamma}_i$ for $1 \leq i \leq 3$.

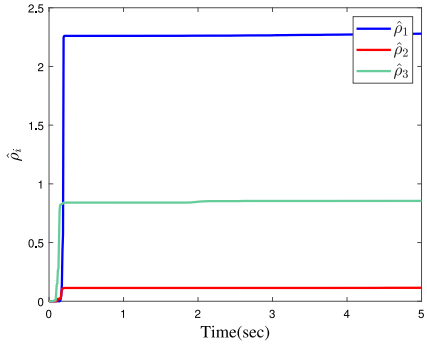


Fig. 6. Parameter estimates $\hat{\rho}_i$ for $1 \leq i \leq 3$.

the desired trajectory $y_d(t)$. Moreover, the boundedness of the states $x_{i,2}$ and the parameter estimates including $\hat{\gamma}_i$, $\hat{\rho}_i$ can be obtained in Figs. 4–6. From these figures, we can clearly obtain that asymptotically consensus tracking can be achieved, and all the signals are bounded.

5. Conclusion

In this paper, a new K-filters based distributed adaptive resilient control scheme is proposed for nonlinear MASs with mismatched unknown parameters against sensor and actuator deception attacks. By constructing a set of new K-filters using compromised output only, which, together with a coordinate transformation involving the attacked output and filter variables, allow a new form of backstepping based distributed resilient adaptive control protocol to be developed. Based on the scheme, the asymptotically consensus output tracking can be achieved, and all the signals in the closed-loop system are globally uniformly bounded.

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